MATH3033: Partial Differential Equations (Spring 2025)
Test 2

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1. [40 marks] Show that the solution of the boundary value problem

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{for } 0 < x < L \\ u(0,t) = 0 = u(L,t) \end{cases}$$

with the initial conditions

$$u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x)$$

is

$$u(x,t) = \sum_n \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}$$

where A_n and B_n satisfy that

$$A_n = \frac{2}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx \qquad \text{and} \qquad B_n = \frac{2}{n\pi c} \int_0^L \psi(x) \sin \frac{n\pi x}{L} dx.$$

[Hint]

$$\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = \frac{L}{2} \delta_{n,m}$$

2. [20marks] Find the Fourier cosine series of the function x

3. [40marks, 10marks each]

(a) Starting with the product rule for $(vu_x)_x$, show the Green's second identity

$$\iiint_D (u\Delta v - v\Delta u) dxdydz = \iint_{\partial D} (u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}) dS$$

(b) The Green function $G(\mathbf{x}, \mathbf{x}_0)$ is defined by the one satisfying $G(\mathbf{x}, \mathbf{x}_0) = G(\mathbf{x}_0, \mathbf{x})$, $\Delta G(\mathbf{x}, \mathbf{x}_0) = \delta(\mathbf{x} - \mathbf{x}_0)$ in $\mathbf{x} \in D$, $G(\mathbf{x}, \mathbf{x}_0) = 0$ on $\mathbf{x} \in \partial D$. Show that the solution of $\Delta u = f$ in D, $u = h$ on ∂D is given by

$$u(\mathbf{x}_0) = \iint_{\partial D} \frac{\partial G(\mathbf{x}, \mathbf{x}_0)}{\partial n} h(\mathbf{x}) dS + \iiint_D G(\mathbf{x}, \mathbf{x}_0) f(\mathbf{x}) d\mathbf{x}$$

(c) Following the definition in (b), show that in 3D the Green's function for a half-space $D = \{z > 0\}$ is

$$G(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{4\pi |\mathbf{x} - \mathbf{x}_0|} + \frac{1}{4\pi |\mathbf{x} - \mathbf{x}^*{}_0|}$$

where $\mathbf{x} = (x, y, z)$ and $\mathbf{x}^* = (x, y, -z)$.

(d) Show that in 3D the solution of Laplace equation

$$\begin{cases} \triangle u = 0, & \text{in } D \\ u = h, & \text{on } \partial D \end{cases} \tag{1}$$

for a halp-space $D = \{z > 0\}$ is represented by

$$u(\mathbf{x}_0) = \iint_{\partial D} \frac{1}{2\pi} \frac{z_0}{|\mathbf{x} - \mathbf{x}^*{}_0|^3} h(\mathbf{x}) dS$$